

CSE 4511: Computer Networks**QUIZ 3****DURATION: 30 MINUTES****WINTER SEMESTER, 2024-2025****FULL MARKS: 20****SET: B****STUDENT ID:**

1. a) Suppose in a custom transport layer protocol named 'FAH', the field to hold the sequence number has 4 bits. What is the minimum and maximum size of the window if it follows the sliding windows technique to send packets?

2
(CO1)
(PO1)

Solution:

Minimum - 1, maximum - 15 or $2^4 - 1$

Rubric:

1. For each correct answer 1

- b) Is it possible to have all 0's in the checksum field? Give proper reasoning

2
(CO1)
(PO1)

Solution:

Yes, if the sender doesn't want to use checksum

Rubric:

1. 1 mark for correct Yes
2. 1 mark for correct reasoning

2. a) What is the main differences between ACK and SACK?

3
(CO1)
(PO1)

Solution:

In Transmission Control Protocol, an ACK (Acknowledgment) is cumulative, meaning it confirms the receipt of all bytes up to a certain sequence number and assumes all previous data has been received correctly. However, it does not provide information about any missing segments beyond that point. In contrast, SACK (Selective Acknowledgment) allows the receiver to explicitly inform the sender about non-contiguous blocks of data that have been successfully received. This enables the sender to retransmit only the missing segments rather than all subsequent data, improving efficiency especially in cases of packet loss.

- b) In TCP we use sequence number in ACK, in SCTP which number is used in SACK? and why?

1 + 2
(CO1)
(PO1)

Solution:

In Stream Control Transmission Protocol, the Transmission Sequence Number (TSN) is used in SACK instead of the byte-based sequence number used in TCP. This is because SCTP operates with message-oriented transmission and supports multiple streams. TSN provides a global sequencing mechanism across the entire association, allowing accurate

acknowledgment, gap detection, and retransmission. Using TSN ensures reliable delivery across all streams, which SSN and SI alone cannot guarantee.

c) How congestion control is achieved in TCP?

4
(CO1)
(PO1)

Solution:

In Transmission Control Protocol, congestion control is achieved by dynamically adjusting the senders transmission rate based on network conditions using the congestion window (cwnd). Initially, TCP uses slow start, where cwnd increases exponentially to quickly utilize available bandwidth. Once a threshold is reached, it enters congestion avoidance, where cwnd grows linearly to prevent sudden congestion. When packet loss is detected (via timeout or duplicate ACKs), TCP reduces the congestion window using fast retransmit and fast recovery mechanisms. This adaptive adjustment ensures efficient data transfer while avoiding network congestion.

3. A UDP segment has the following fields (all values are in decimal):

6
(CO1)
(PO2)

- Source IP: 255.200.10.5
- Destination IP: 128.250.100.25
- Protocol: 17
- UDP Length: 13 bytes
- Source Port: 60000
- Destination Port: 61000
- Data bytes: (255, 1, 128, 64, 32)

Calculate the UDP checksum.

Solution:

Given (decimal):

- Source IP: 255.200.10.5
- Destination IP: 128.250.100.25
- Protocol: 17
- UDP Length: 13 bytes
- Source Port: 60000
- Destination Port: 61000
- Data bytes: (255, 1, 128, 64, 32)

Step 1: 16-bit binary words

Src IP = 1111111111001000, 0000101000000101

Dst IP = 1000000011111010, 0110010000011001

Zero + Proto = 00000000000010001

UDP Length = 0000000000001101

Src Port = 1110101001100000

Dst Port = 1110111001001000

Data = 1111111100000001, 1000000001000000, 0010000000000000

Step 2: 1's complement addition

Final Sum = 0110011011111001

Step 3: Complement 1001100100000110